

## 2.2 - Derivatives

**Goals:** A shorter than normal worksheet with a few extras on derivatives

**Before Class Videos:** Optional tricky example from last class's content: 2.1.4

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**Definition:**

The derivative of  $f(x)$  at  $c$  is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

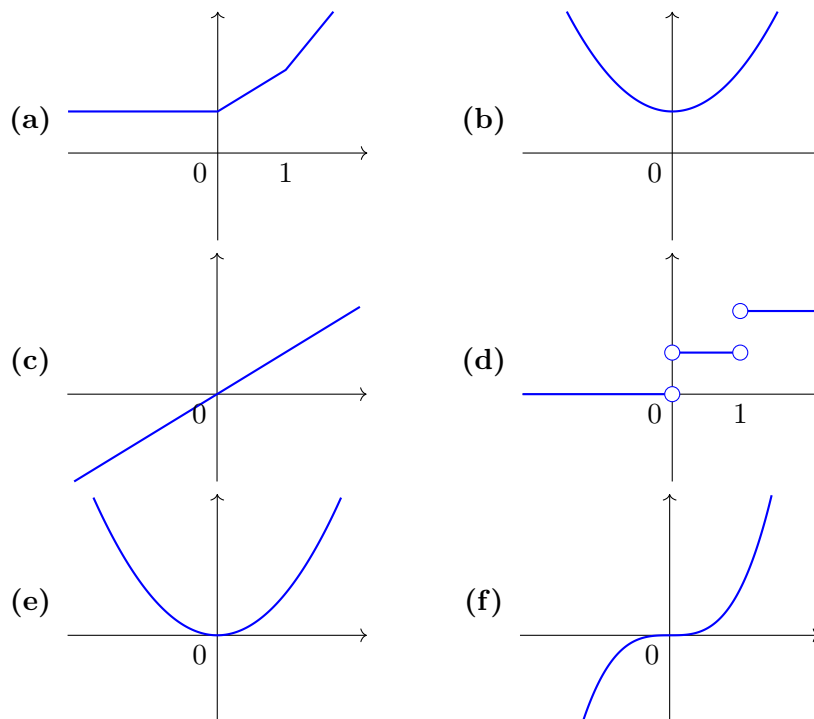
provided this limit exists. If it does, we say  $f$  is differentiable at  $c$ .

**Definition:**

An alternate formula for the derivative of  $f(x)$  at  $c$  is

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

**Example 1:** The following six graphs consists of three pairs of functions and their derivative. Match them up.



**Theorem:**

Differentiable  $\implies$  Continuous.

Note: This worksheet is deliberately short. I anticipate this class being a bit behind and finishing up prior material. If I am wrong I will edit it and add to it. The next worksheet is a more normal length.

**Suggested after class problems** from APEX: Section 2.2 15,17